IonVe
ELECTROCHEMICAL REACTORS

Long-Duration Industrial Energy Storage

IonVe develops redox flow battery systems targeting 4–12 hour storage for heavy industry. Architecture redesigned from cell level. Design-target phase.

TRL 4-5 · Pre-incorporation · PoC target: June 2026

Why long-duration storage, and why now?

Industrial demand charges are the real cost driver

Heavy industry sites (steel, foundries, cement, recycling) face electricity bills of €1–8M/year. 30–40% of that cost comes from peak-demand charges — consumption spikes that last just a few hours per day.

Lithium-ion is not cost-effective beyond 4–6 hours

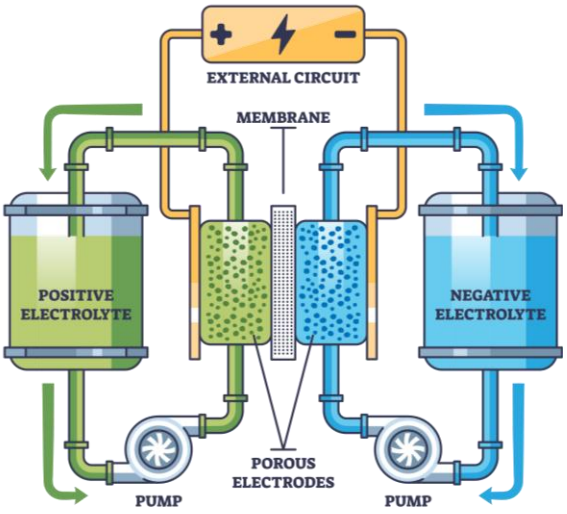
At longer durations, Li-ion requires replacement every 10–12 years. Over a 20–30 year industrial lifespan, total cost of ownership is significantly higher than a flow battery — even at current flow battery prices.

Existing flow batteries are still too expensive

Current VRFB products are priced at €450–600/kWh installed. High component count, machined graphite parts, and site-specific engineering drive costs. No standard catalogue product exists today.

What is a flow battery?

Think of a liquid-filled tank that produces electricity as the fluid passes through an electrochemical cell.
More tank volume = more hours of autonomy. Unlike lithium-ion, the chemistry does not degrade over time — capacity depends only on tank size.



Lithium-ion

Degrades with cycles. Replacement every 10 yr. Best for <4 h.

Flow Battery

30+ yr lifespan, no capacity loss. Designed for 4–12 h daily cycling.

IonVe architecture — what we are targeting

All figures below are design targets from modelling based on ongoing testing and real case studies.

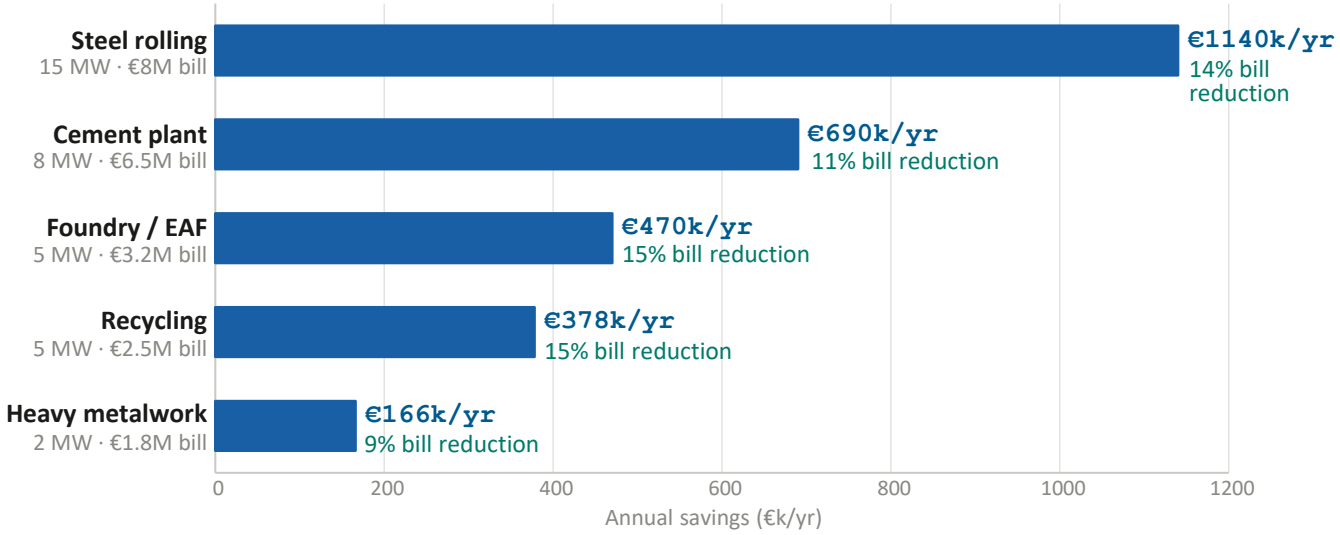
297 €/kWh
Target installed cost (6 h)
−40% vs. current VRFB

>30 years
Operating life
No degradation

~5 yr
Simple payback (foundry
500 kW / 3 MWh reference)

50 MW
Max scale — same
architecture, no re-engineering

Modelled annual savings by sector (design-target system)



Savings model: ~65% from demand-charge reduction (peak shaving), ~35% from time-of-use arbitrage. Site-specific validation pending. IonVe recommends Li-ion for durations <4 h.

30-year TCO comparison — 500 kW / 3 MWh

ROADMAP & NEXT STEPS

1. Proof of Concept · Jan–Jun 2026 · TRL 3→5 ← We are here

MK1/2 hydraulic & electrical tests · 3 go/no-go criteria · Patent filings (3 applications)

2. Functional Prototype · Jul–Dec 2026 · TRL 5→6

2–3 prototypes · 250–500 cycle endurance · Independent lab validation

IonVe MK1 prototype stack modules in test

3. Industrial Pilot · 2027 · TRL 6→7

2–3 co-development pilots · Supply-chain qualification · Series A fundraise

4. Commercial Scale · 2028–2030 · TRL 7→9

15–30 systems/yr, 5 countries · EPC channel partnerships · Energy-as-a-Service model

How we work with partners

01 Site-Specific Screening — No cost, no obligation

We model annual savings from your 15-min interval load data and electricity contract. Output: site report with NPV, payback, and sensitivity within 2 weeks of data receipt.

02 Access to Test Data — Under NDA, on request

We share electrochemical results and stack performance data as milestones are completed. MK1 hydraulics done; MK2 characterisation in progress.

03 Co-Development Pilot — Post-PoC, from Q3 2026

2–3 pilot installations at cost-plus pricing in exchange for real operating data. Independent performance verification. Right of first negotiation.

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Visit our website for more information!

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All performance and cost figures are design targets. Validation in progress.